



Progress Report: Formal Verification of the 5G AKA Protocol

Cyber Security Lab I

Nour Hmeedan

This report summarizes the progress made until the end of April for Cyber Lab 1. The work focuses on the formal verification of the 5G AKA protocol using the Tamarin Prover. The objective of this lab is to model the security aspects of communication protocols, particularly sequence number handling, key agreement, and authentication in a confidential data processing architecture.

The 5G AKA model used is based on the publicly available reference implementation from the Tamarin-Prover GitHub repository. After downloading the model and installing the required dependencies, including OCaml, Stack, and Maude, the Tamarin Prover was successfully compiled and executed under WSL2 with Ubuntu 24.04.

I have started working on proving the security lemmas associated with the 5G AKA protocol. The lemmas cover aspects like sequence number invariance, authentication commitments, and the correctness of the key derivation processes. The Tamarin formal prover was used to interactively inspect, apply, and validate the proof scripts, including inductive and automatic reasoning tactics.

Currently, multiple lemmas have been successfully verified and the derivation checks are actively being resolved. The model includes key rules such as `add_subscription`, `init_homeNet`, and `seaf_key_confirmation_check`. Ongoing efforts are focused on finalizing the remaining proofs and preparing a comprehensive implementation report. The future steps include formulating and proving a new lemma related to the security properties of the protocol.